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Two New Species of *Aira* (Poaceae) from the Iberian Peninsula and the Balearic Islands

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Abstract*—**Two new diploid species, ***Aira minoricensis and ***Aira hercynica,*** are described and illustrated, along with chromosome counts, risk assessment, distribution and habitat, phenology, and comparisons with morphologically similar species. A comparative table and a key for the species of *Aira* for the Iberian Peninsula and the Balearic Islands are provided to assist in the identification of these overlooked species, and their relationships to other taxa are discussed.

Resumen*—*Aira minoricensis y ***Aira hercynica*** son dos nuevas especies diploides que se describen y se ilustran en el presente trabajo. Se aportan recuentos cromosómicos, estado de conservación, distribución y hábitat, fenología y análisis morfológico de estos taxones y de las especies más próximas a ellos. Se incluye una tabla comparativa y una clave de las especies de *Aira* de la Península Ibérica y las Islas Baleares para facilitar la identificación de las nuevas especies y también se discute su relación con otros taxones relacionados.

***Keywords*—**Endemism, Mediterranean region, taxonomy.

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Linnaeus (1753) described the genus *Aira* L. as a heterogeneous complex of 14 species of grasses characterized by spikelets with two flowers. As knowledge improved, the wide Linnaean generic concept of genus *Aira* was narrowed to a more restricted circumscription of the genus (Gaudin 1811; Palisot de Beauvois 1812; Dumortier 1824). Nowadays those taxa are classified in 11 distinct genera, according to Soreng et al. (2017). At the end of the 19th century Ascherson and Graebner (1899, sub *Aera*) included in different sections and subsections of genus *Aira*, 10 European species, five of which are presently classified as genus *Aira*. The others are included in genera *Avenella* Drejer, *Aristavena* F.Albers & Butzin, or *Deschampsia* P.Beauv. The current use of the name *Aira* was retained by conservation, adopting *Aira praecox* L. as the conserved type (Jarvis 1992). The modern concept of genus *Aira* was well defined by Rouy (1913), who circumscribed the genus to include the six Mediterranean species recognized to date, and established four sections within the genus: sect. *Pycnaira* Rouy nom. illeg. including the conserved type of the genus, monotypic with *A. praecox*; sect. *Euaira* nom. inval., including taxa with moderately lax panicles (e.g. *A. caryophyllea* L.); *Trichodaira* (Aschers. & Graebn.) Rouy, taxa with wide, very lax panicles (e.g. *A. tenorei* Guss.), and *Airastrum* Rouy, monotypic with *A. provincialis* Jordan.

Aira is a relatively small genus with eight (Clayton & Renvoize 1986; Watson and Dallwitz 1992) to 10 species (Albers 1980c), within the tribe *Poeae* R.Br., subtribe *Airinae* Fries, according to Soreng et al. (2017). This genus diversified along the Mediterranean Basin and includes only slender, annual plants, which usually occur in dry open habitats (Albers 1980c; Tutin 1980). To our knowledge, there is no recent taxonomical review or phylogeny of *Aira*. However, the subtribe *Airinae* was long investigated by Albers (1973, 1975, 1978a, 1978b, 1978c, 1979, 1980a, 1980b, 1980c), Albers and Albers (1973), and Albers and Butzin (1977), who contributed to the karyology and other taxonomic aspects of the genus *Aira*. Albers (1980c) distinguished 12 taxa in the genus, 10 species and two subspecies, six diploid ($2n = 14$) and six tetraploid ($2n = 28$). Quintanar et al. (2007) studied the subtribe *Airinae,* and based on nuclear and plastid molecular markers, stated that *Aira* is sister to *Corynephorus* P.Beauv., *Deschampsia*, and *Periballia* Trin., including a single sample of *A. cupaniana* Guss.

Willkomm (1861) listed seven *Aira* species for Spain, six of which are currently accepted: *A. praecox*, *A. multiculmis* Dumort., *A. cupaniana*, *A. caryophyllea*, *A. elegans* Roem. & Schult., and *A. tenorei*. *Aira lendigera* Lag. was also listed by Willkomm (1861), which is currently regarded a taxonomic synonym of *Molineriella laevis* (Brot.) Rouy according to Cebrino and Romero-Zarco (2017).

Paunero (1946, 1957) studied morphologically the Spanish taxa of genus *Aira* based on the specimens preserved in the herbarium of Real Jardín Botánico de Madrid (MA). She recognised six species: *A. caryophyllea*, *A. cupaniana*, *A. elegans*, *A. praecox*, *A. tenorei*, and *A. uniaristata* Lag. & Rodr. Within *A. caryophyllea* she distinguishes two subspecies: subsp. *caryophyllea* and subsp. *multiculmis* (Dumort.) Bonnier & Layens. She also clarified the true identity of *A. uniaristata*, usually misidentified by other authors as *A. cupaniana*. The diagnostic characters used to evaluate the taxa identity are panicle shape and density, pedicel/spikelet length, shape of the thickened pedicel apex, spikelet size, and anther length (Paunero 1957; Tutin 1980).

In the framework of the ongoing project *Flora Iberica*, an exhaustive taxonomic review was carried out with the specimens conserved in the principal herbaria of the Iberian Peninsula, in Spain and Portugal (Sáez et al. in prep.). As a result of this study two new taxonomic entities were detected, one endemic to Menorca Island (named hereafter *A. minoricensis*) and another endemic to the Iberian Peninsula (*A. hercynica*). The aim of present study is to characterize both entities and add precise data to distinguish them from their

close relatives, determine chromosome number, and distributional area.

Following the infrageneric classification mentioned above, which has not been adopted in modern floras, four groups without taxonomic rank can be recognized in the genus for practical purposes. The morphology of the panicles and the size of the spikelet are the basic characters to define these four groups. The first species group, morphologically similar to *A. caryophyllea* or *A. cupaniana*, is characterized by lax, not diffuse, inflorescences, and spikelets with variable size. Other species related with this caryophyllean group are *A. armoricana* F.Albers, *A. multiculmis*, commonly treated as subordinate to *A. caryophyllea* at subspecies level, and *A. uniaristata*. A second group, here called the *A. elegans* species group, includes three species with very lax and diffuse inflorescences and small spikelets, up to 2.5 mm: *A. elegans*, *A. tenorei*, and *A. byzantina* F.Albers. A third monospecific group composed of *A. provincialis* shows intermediate characteristics between the two previous groups. And the last one, also monospecific, comprises only *A. praecox*, with large spikelets (3–3.8 mm), the single known *Aira* species with a rather dense inflorescence and short branches that remain erect, adpressed to the rachis, even in anthesis. This work focussed on the first group, to which the two new detected taxa show greatest similarity.

Materials and Methods

Morphological observations and measurements of vegetative and reproductive parameters were undertaken on plant material kept in the herbaria BC, COFC, G, HVR, LISI, LISU, MA, MAF, SANT, SEV, and UNEX (acronyms according to Thiers 2018), and is available as a supplemental file at the Dryad Digital Repository (Sáez et al. 2019). For the morphological observations, an Olympus SZX12 and a Leica S6D binocular microscopes were used. A total of 24 morphological characters recognized as taxonomically relevant by authors dealing with the taxonomy of *Aira* (Paunero 1957; Tutin 1980; Pignatti 1982; Romero-Zarco 1987, 2002; Bolòs and Vigo 2001), as well as our own observations, were selected, of which 22 were quantitative (including one calculated ratio) and two semiquantitative (Table 1); these characters were scored in 134 specimens. A list of all the studied specimens is provided in the supplemental file (Sáez et al. 2019). Well-developed specimens were selected for the morphometric analyses. Well-developed spikelets were selected from the medium level of the panicle. The multivariate ordination of these individuals based on the measured variables was assessed with a principal component analysis (PCA) using the SPSS v. 22.0 software (IBM Corp. 2013). Before PCA, some exploratory data analysis, including normality test and KMO test, were conducted with SPSS v. 22.0, PAST 3.19 (Hammer et al. 2001) and Factor 10.8.04 (Lorenzo-Seva and Ferrando 2006), respectively. Furthermore, a cluster analysis was performed by the paired group algorithm (UPGMA, unweighted pairgroup method with arithmetic mean, Michener and Sokal 1957) and Gower similarity index (Gower 1971) using PAST 3.19.

For the chromosome counts, seeds were collected in Huelva, Gibraleón, (SEV 287018 ex duplis) and Menorca, Alzinar d'Alforí (SEV 287206). Root tips were pretreated in 2 mM 8-hydroxyquinoline solution for 3 h from seeds germinated in Petri dishes or from plants growing in pots. This material was fixed in a solution of absolute ethanol and glacial acetic acid (3:1) for 24 h at room temperature, and stored at -20°C. Subsequently, root-tip meristems were excised, washed with distilled water, stained with alcoholic hydrochloric-acid carmine (Snow 1963) for 24 h and squashed on a drop of 45% acetic acid solution. Slides were observed with a Leica DM 2000 microscope and the best metaphase plates were photographed with a Leica DFC 295 camera.

Conservation status was assessed according to IUCN (2012, 2017). We calculated the area of occupancy (AOO) and extent of occurrence (EOO) using the geospatial conservation assessment tool GeoCAT (Bachman et al. 2011).

A Phenom Pro low vacuum scanning electron microscope was used for the photographs of the apical thickening of the pedicels.

Table 1. Morphological variables used in the morphometric analyses.

Morphological characters	Type
Panicle length (mm)	Quantitative
Number of panicle nodes	Quantitative
Panicle branch length (mm)	Quantitative
Maximum spikelet number	Quantitative
Minimum peduncle length (mm)	Quantitative
Maximum peduncle length (mm)	Quantitative
Peduncle width (1–3): 1: <0.09 mm; 2: 0.09–0.1 mm; 3: >0.1 mm	Semiquantitative
Apical pedicel thickening (1–3): 1) with a conspicuous annular widening; 2) claviform; 3) conical-subcylindrical	Semiquantitative
Lower glume length (mm)	Quantitative
Lower glume width (mm)	Quantitative
Upper glume length (mm)	Quantitative
Upper glume width (mm)	Quantitative
Callus hairs length (mm)	Quantitative
Lemma length (mm)	Quantitative
Lemma apex length (mm)	Quantitative
Palea length (mm)	Quantitative
Awn number	Quantitative
Basal portion awn length (mm)	Quantitative
Ratio of basal portion awn length / lemma length	Quantitative
Column length (mm)	Quantitative
Setae length (mm)	Quantitative
Awn length (mm)	Quantitative
Anther length (mm)	Quantitative
Number of ramification order	Quantitative

Results

Morphological Analyses—Exploratory data analysis showed departures from normality for nearly all the analyzed variables; however, lack of normality is argued to be tolerable if the analysis

Table 2. Principal Component Matrix showing the contribution of each morphological character to the two principal components (PC1 and PC2) for *Aira caryophyllea* subsp. *caryophyllea*, *A. caryophyllea* subsp. *multiculmis*, *A. cupaniana*, *A. hercynica*, *A. minoricensis*, and *A. uniaristata*. Traits with higher scores are highlighted in bold.

	Components	
Morphological characters	1	2
Panicle length	0.12	**0.80**
Number of panicle nodes	**-0.46**	0.53
Number of ramification order	−0.37	0.42
Panicle branch length	0.21	**0.73**
Maximum spikelet number	−0.30	0.18
Minimum peduncle length	0.68	0.27
Maximum peduncle length	0.73	0.22
Peduncle width	0.38	−0.39
Apical pedicel thickening	0.27	−0.31
Lower glume length	**0.92**	−0.06
Lower glume width	0.86	−0.10
Upper glume length	**0.95**	−0.01
Upper glume width	0.86	−0.16
Callus hairs length	0.87	−0.10
Lemma length	**0.96**	−0.03
Lemma apex length	0.25	−0.53
Palea length	**0.92**	0.19
Awn number	−0.09	**-0.65**
Basal portion awn length	0.54	−0.52
Ratio basal portion awn length / lemma length	−0.41	−0.58
Column length	0.91	0.20
Setae length	0.80	−0.09
Awn length	0.91	0.02
Anther length	0.47	**0.63**

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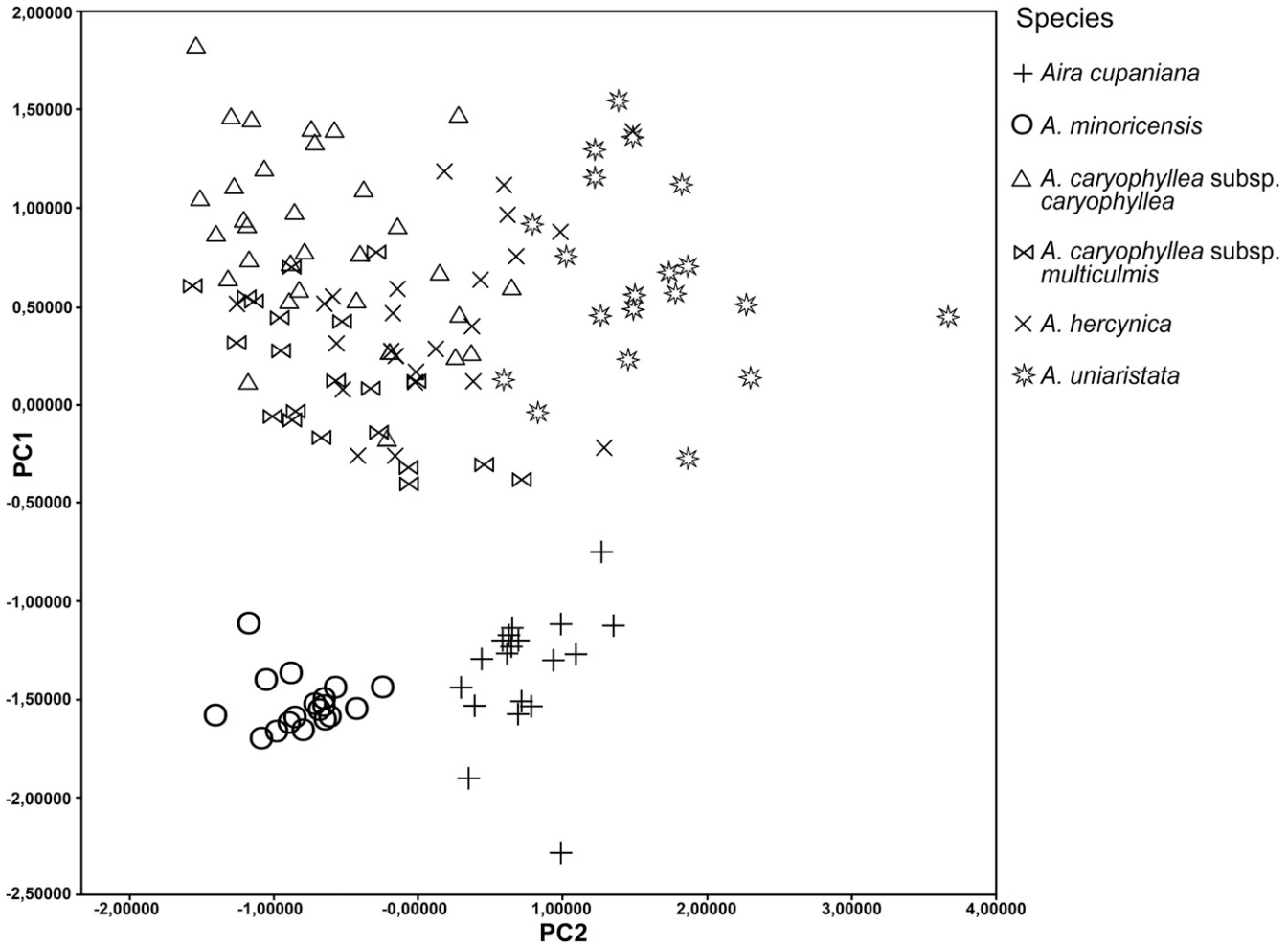


FIG. 1. Ordination of principal component analysis for morphological characters for *Aira caryophyllea* subsp. *caryophyllea*, *A. caryophyllea* subsp. *multiculmis*, *A. cupaniana*, *A. hercynica*, *A. minoricensis*, and *A. uniaristata*. PC1 = Principal Component 1. PC2 = Principal Component 2.

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is conducted for descriptive purposes (Gauch 1982). Furthermore, the KMO test yielded a value of 0.68, which indicates that our data is suitable to conduct a PCA analysis.

The PCA analysis (Table 2; Fig. 1), explained the 60.01% of total variance for the first two principal components (PC) (43.82% for PC1 and 16.19% for PC2 axes). Characters contributing most to the first PC axis are glume length (length for upper and lower glumes), lemma and palea lengths, and awn and column lengths (positively), whereas number of panicle nodes contributes negatively. Among the characters contributing most to the second PC axis are panicle length, panicle branches length, and anther length as positive values, and awn number as the most negative one. According to the PCA ordination (Fig. 1) *A. minoricensis* appears the most closely related to *A. cupaniana*. Specimens from both species could be mainly separated along the second PC axis, whereas they overlapped along the first one. On the contrary, the first axis separates clearly *A. minoricensis* from *A. caryophyllea* (in broad sense). Overlapping between *A. caryophyllea* subsp. *caryophyllea* and *A. caryophyllea* subsp. *multiculmis* supports the infraspecific arrangement proposed by Stace (2010). The PCA also shows overlap between *A. caryophyllea* in broad sense and *A. hercynica,* whereas *A. uniaristata* separates from *A. caryophyllea* and *A. hercynica* mainly along the second PC axis.

The UPGMA dendrogram (Fig. 2) revealed two well-delimited main clusters, one formed by *A. minoricensis* and *A. cupaniana* and the other formed by the remaining species, both clusters with irrelevant mixing among species. Within this second cluster all the *A. hercynica* samples formed another separate cluster. Most remarkable intermixing occurs between *A. caryophyllea* subsp. *caryophyllea* and *A. caryophyllea* subsp. *multiculmis*.

Comparative Morphology—Table 3 summarizes several morphological characters of the inflorescence and the spikelet of *A. minoricensis* and its most related taxa. Within these results, the shape of the panicle and pedicels (Fig. 3) and spikelet and anther length are the most significant characters to distinguish among the taxa. In the case of *A. hercynica,* the diploid chromosome number and large anthers clearly point to *A. uniaristata* as its most similar species (Table 4).

Chromosome Numbers—The studied material of *A. minoricensis* from the type locality (SEV 287206) shows a chromosome number $2n = 14$ (Fig. 4A). The same number, $2n = 14$, was obtained in several seedlings of *A. hercynica* from Gibraleón, Huelva (SEV 287018, Fig. 4B).

DISCUSSION

The morphological analyses performed are congruent with Albers´concept of species (Albers 1980b), and support the taxonomic treatment followed for the new described taxa. *Aira minoricensis* is isolated due to its ecology and its insular

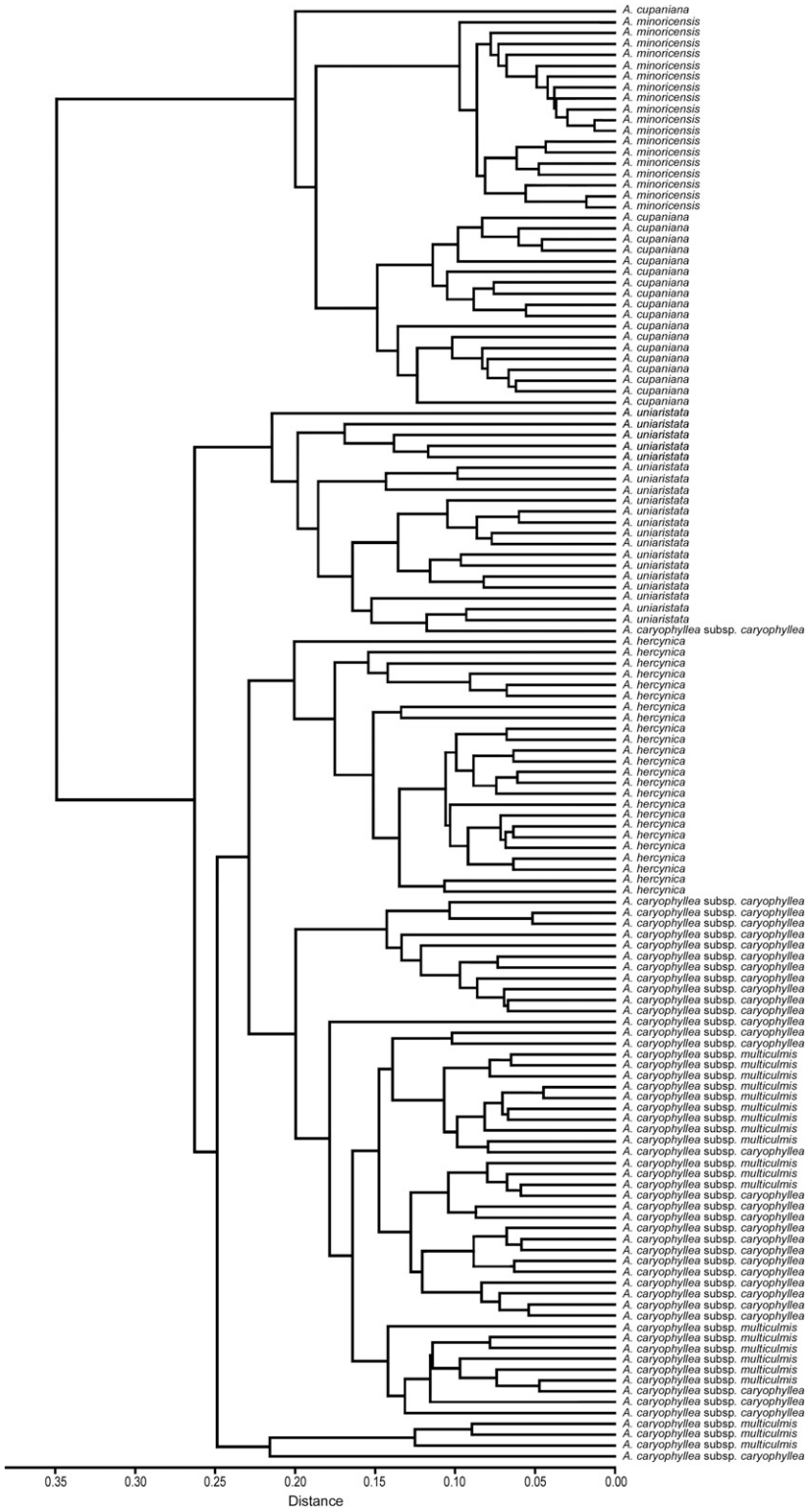


FIG. 2. Hierarchical clustering of individuals of *Aira caryophyllea* subsp. *caryophyllea*, *A. caryophyllea* subsp. *multiculmis*, *A. cupaniana*, *A. hercynica*, *A. minoricensis* and *A. uniaristata* using paired group algorithm (UPGMA) and Gower similarity index.

distribution. However, it is morphologically related to the diploid ($2n = 14$) *A. cupaniana*, a Mediterranean species also present on Menorca Island, sharing with it some flower features (glumes usually short, callus glabrous or with short hairs). Nevertheless, *A. minoricensis* is easily distinguished from *A. cupaniana* by its narrow and more or less dense panicles with erect branches. Moreover, *A. cupaniana* differs from the new species by its pedicels abruptly thickening at the apex, with a conspicuous annular widening (Fig. 3B) and lower floret sometimes unawned.

Another somewhat related species to *A. minoricensis* is *A. caryophyllea*, a tetraploid ($2n = 28$) plant (Pohl and Davidse

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TABLE 3. Morphological comparison of *Aira minoricensis* and related species.

Character	*A. caryophyllea* subsp. *caryophyllea*	*A. caryophyllea* subsp. *multiculmis*	*A. cupaniana*	*A. minoricensis*
Panicle	Wide	Wide	Wide	Narrow
Pedicel shape	Without a conspicuous annular widening	Without a conspicuous annular widening	With a conspicuous annular widening	Without a conspicuous annular widening
Pedicel length (mm)	1–7.5(–10)	0.8–6.0	0.5–5.7	0.5–2.7
Spikelet length (mm)	(2.6–)2.7–3.5	(2.2–)2.3–2.7(–2.8)	1.6–2.4	1.9–2.2
Glume shape	Acute	Subobtuse or acute	Obtuse or subobtuse	Acute
Glume length (mm)	2.6–3.5	2.2–2.8	1.5–2.4	1.8–2.2
Lemma length (mm)	2.0–2.8	1.9–2.4	1.2–2.2	1.2–1.5
Palea length (mm)	1.4–1.9	1.2–1.7	0.9–1.7	0.8–1.2
Awn length (mm)	3.0–4.3	2.9–3.8	1.6–3.2	1.9–2.3
Callus hairs length (mm)	0.3–0.6	0.3–0.5	0.1–0.3	ca. 0.1
Anther length (mm)	0.20–0.50(–0.60)	0.20–0.50	0.20–0.30(–0.50)	0.28–0.37

1971; Albers 1980c; Kirschner et al. 1982) (diploid sporophytic counts obtained from plants collected in central Africa (Morton 1993) and Central Europe (Krahulcová 2003) require verification) which has been regarded as a polymorphic species (Paunero 1957). Both subspecies of *A. caryophyllea* have longer spikelets and pedicels than *A. minoricensis* (Table 3; Tutin 1980). *Aira praecox*, somewhat similar in its habit, differs from the new species by its longer glumes (3–3.6 mm), longer lemma (2.6–3 mm), and longer awns (4.1–4.6 mm) among other characters (Sáez et al. in prep.).

Aira hercynica is close to the diploid *A. uniaristata* ($2n = 14$), endemic to Betic and Riff mountains (Romero-Zarco 2002), differing in having apical thickening of pedicels that is conical-subcylindrical (Fig. 3C) and approximately twice as long as wide (vs. apical thickening claviform, approximately as long as wide; Fig. 3D), and shorter anthers (Table 4). Taxonomic relationships of *A. hercynica* with other species of the genus are remote. However, the herbarium material of this species has been misidentified as *A. caryophyllea*, *A. uniaristata,* or *A. elegans*. A few mixed populations have been detected only with *A. caryophyllea*, but no intermediate plants have been detected.

In addition to the species previously known from the Spanish flora, two other species have been described from western Europe and the Mediterranean Basin by Albers (1979): *Aira armoricana* and *A. byzantina*. The first resembles *A. caryophyllea* subsp. *multiculmis*, from which it differs mainly by the larger size of its spikelets (3–3.5 mm). *Aira armoricana* is a tetraploid ($2n = 28$) species with relatively small anthers (0.3–0.6 mm). It is endemic to northwestern France and Cornwall in southwestern Britain. *Aira byzantina* is a diploid ($2n = 14$) species morphologically close to *A. elegans* and is known from southern Balkans and western Asia Minor. *Aira byzantina* shows a very lax and diffuse panicle, very different from the two new species described below. Of these two species only *Aira armoricana* belongs to the group of species investigated in this paper, but it differs significantly from the two new entities described here. *Aira hercynica* and *A. minoricensis* present smaller spikelets than *A. armoricana* (see description below; Tables 3, 4). Furthermore, *A. armoricana* can be distinguished from *A. hercynica* by its relatively small anthers. *Aira minoricensis* clearly differs from *A. armoricana* by its dense panicle.

Our findings resulting from these studies lead us to the conclusion that we are faced with two *Aira* entities sufficiently distinct to merit their recognition as new taxa, for which we believe the rank of species to be the most appropriate. The phylogenetic relationships of the new species within the *A. caryophyllea* group will be studied in the near future.

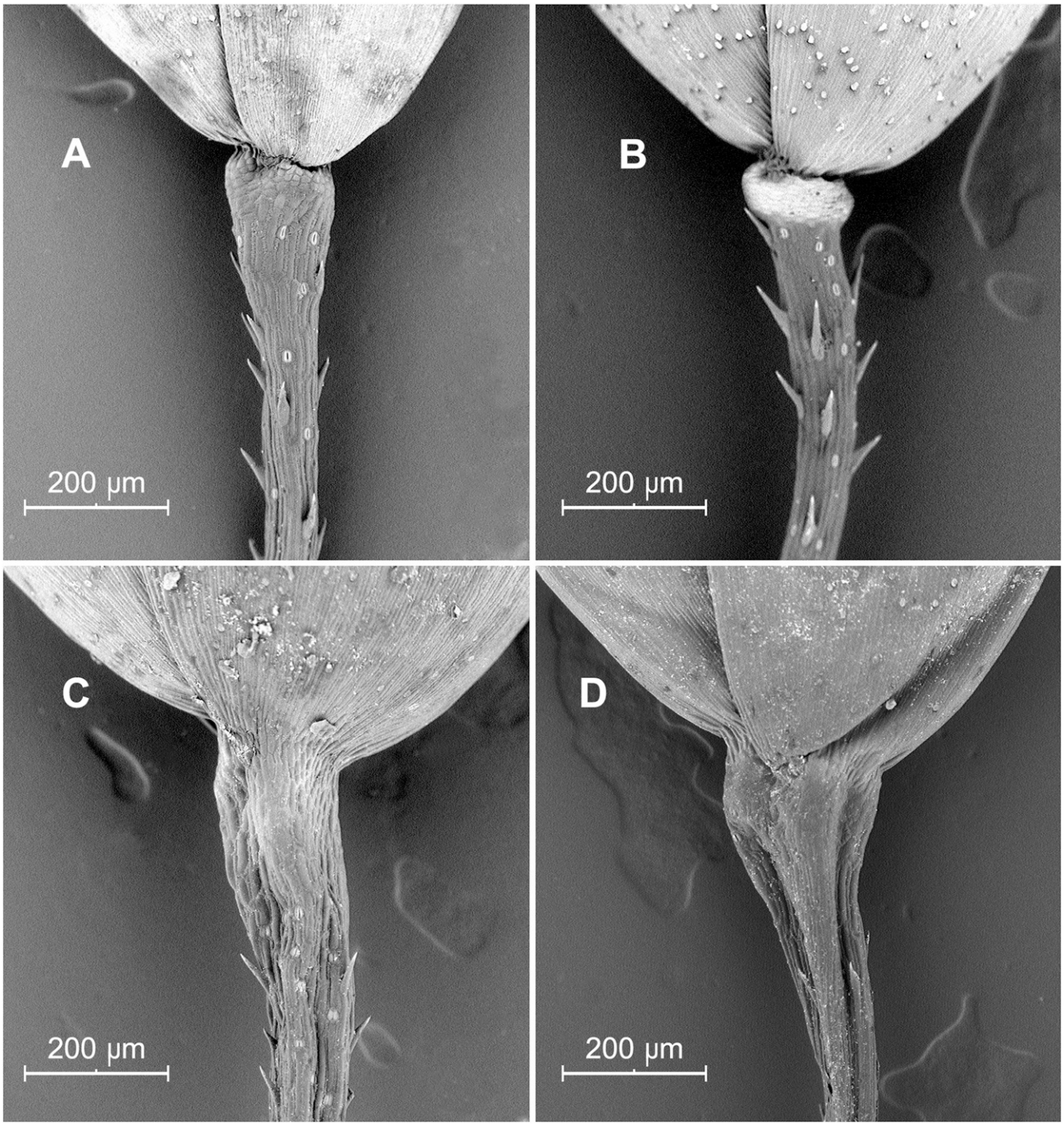
It seems surprising that these two new species have gone unnoticed to date. Its small distributional area can explain the case of *A. minoricensis*, but that is not the case of *A. hercynica*, a widely distributed species in the Iberian Peninsula. Reviewing the specimens determined by Paunero (1957) as *A. caryophyllea*, we found some vouchers of the new species *A. hercynica* misidentified. Probably she could not distinguish them from *A. caryophyllea* because these plants were in fruit stage and anthers were not available. On the other hand, Albers (1978a) did not include in his studies specimens from the localities where *A. hercynica* or *A. minoricensis* have been found. Certainly, the anther size of *A. hercynica* or the contracted panicle of *A. minoricensis* would have drawn his attention.

Our results on chromosome numbers are limited to one population of each new species, but according to Albers (1978a), chromosome numbers in *Aira* are constant at species level. In the studied material (Albers 1978a: 133–159) he included two Spanish localities of *A. cupaniana* ($2n = 14$); three other vouchers of *A. praecox* ($2n = 14$) from Portugal; three diploid counts for *A. uniaristata* from Cádiz province (Spain); two tetraploid counts ($2n = 28$) for *A. multiculmis* from Portugal, and two Spanish vouchers of tetraploid ($2n = 28$) identified by the author as *A. caryophyllea* subsp. *edouardii*, nom. inval., a form with spikelets about 3.5 mm long also known as *A. caryophyllea* f. *edouardii* Maire. Albers' list of studied material does not include typical *A. caryophyllea* subsp. *caryophyllea* from the Iberian Peninsula, but he cited six counts from other authors for Portuguese localities, five of them tetraploid ($2n = 28$) and only one diploid ($2n = 14$), considered by Albers (1978a, op. cit. 61) an error due to a misidentification of the specimen. This material was collected in Azeitão, near Setúbal in the historical province of Estremadura (Queirós 1973), an area where the new diploid *A. hercynica* is very frequent (see Supplemental File; Sáez et al. 2019). Moreover, a meiotic chromosome count ($n = 7$) provided by Devesa et al. (1990) for *A. caryophyllea* subsp. *caryophyllea* from Badajoz (SW Spain) belongs to *A. hercynica,* according to the identity of the voucher specimen (UNEX 5961). Therefore, we can assume with little doubt that *A. hercynica* is a diploid species. In the case of *A. minoricensis*, there is no karyological data available to compare.

TAXONOMIC TREATMENT

Aira minoricensis P.Fraga, Romero Zarco & L.Sáez, sp. nov. TYPE: SPAIN. Balearic Islands, Minorca: Ciutadella de Menorca, Alzinar d'Alforí, 60 m a.s.l., communities of annual plants in stabilized sand dunes [40.035°, +3.971°], 15 Apr 2014, *P. Fraga & L. Sáez LS 7475* (holotype: SEV 287859!; isotypes: SEV 287060!, BC!).

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FIG. 3. Scanning electron micrographs of pedicels. A. *Aira minoricensis* from isotype, SEV 287060. B. *Aira cupaniana* from Huelva, Parque Moret, SEV 287021. C. *Aira hercynica* from Huelva, Carboneras, SEV 49903. D. *Aira uniaristata* from Cádiz, Los Barrios, SEV 102302.

Annual. **Fertile stems** 7–29 cm long, erect. **Leaf blade** 15–42 × 0.5–1.5 mm, usually convolute. **Ligule** 3–5.5 mm long, oblong-lanceolate, acute to subacute. **Panicle** 2–6 × 0.5–0.7(–1) cm, narrow, usually dense, even in fruiting stage; panicle branches up to 3.6 cm long, erect even in anthesis; pedicel 0.5–2.7 mm long, (0.2–)0.3–1.2(–1.4) times as long as spikelet, straight (sometimes slightly flexuose) not abruptly thickened at apex; apical thickening of pedicel claviform to conical-subcylindrical, approximately twice as long as wide. **Spikelets** 1.9–2.2 mm long, with 2 hermaphrodite awned florets, laterally compressed. **Glumes** equal to slightly unequal, ovate-lanceolate, acute (sometimes subacute), scabridulous on keel, hyaline to stramineous, shiny; lower glume 1.9–2.2 × 0.9–1.2 mm, 1-veined; upper glume 1.8–2.1 × 0.9–1.1 mm, 3-veined. **Lemma** 1.2–1.5 × 0.4–0.5 mm, bifid at

TABLE 4. Morphological comparison of *A. hercynica* and *A. uniaristata*.

Character	*A. hercynica*	*A. uniaristata*
Apical pedicel thickening	Conical-subcylindrical	Claviform
Awn length (mm)	2.2–3.7	2.8–4.3
Anther length (mm)	(0.6–)0.7–0.8(–0.9)	1.0–1.6

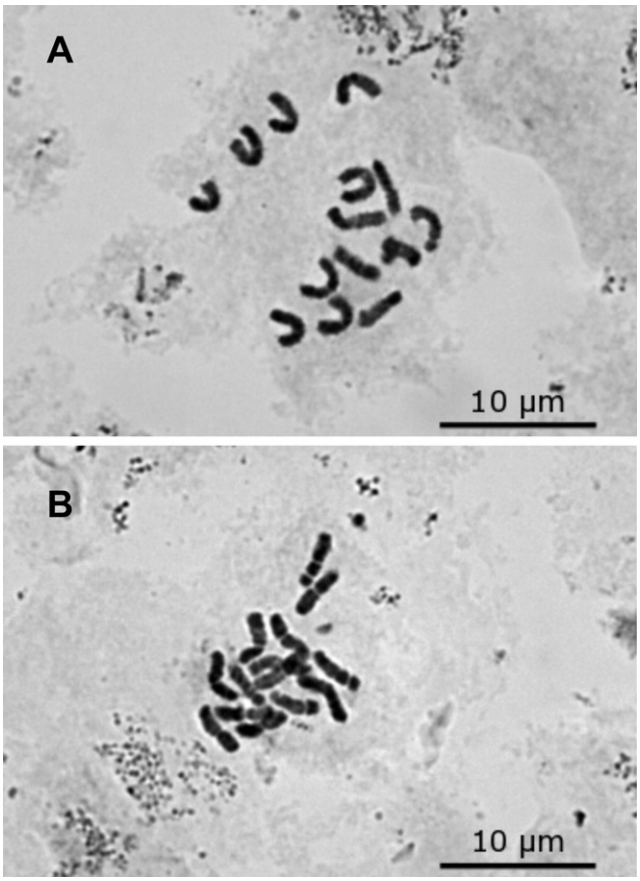


FIG. 4. A. Mitotic metaphase chromosomes of *Aira minoricensis* (SEV 287206). B. Mitotic anaphase chromosomes (partial) of *Aira hercynica* (SEV 287018).

apex, yellowish to red-brown, with a dorsal geniculate awn 1.9–2.3 mm long, arising from lower 1/3–1/2 of lemma. **Callus** usually glabrous, sometimes with hairs up to 0.1 mm long. **Palea** 0.8–1.1 × 0.3–0.5 mm, shortly bifid at apex. **Lodicules** ca. 0.3 mm long, lanceolate, entire. **Anthers** 0.28–0.33(–0.37) mm long. **Caryopsis** 0.8–0.9 × ca. 0.3 mm, oblong-elliptic, longitudinally sulcate on adaxial face, glabrous. Figures 3A, 5.

***Distribution and Habitat*—**At present, *A. minoricensis* is known from few localities in north-western Minorca (Fig. 6). It grows from 15 to 100 m a. s. l. on sandy soils from stabilized maritime dunes, mixed with siliceous materials of Triassic origin. The new species occurs in ephemeral therophyte pastures.

***Etymology*—**The specific epithet refers to Minorca Island, where this species has been discovered.

***Phenology*—**The flowering period occurs from mid-March to the end of April, and fruiting from mid-April to the end of May.

***Conservation*—**The species is known from two localities. *Aira minoricensis* has an extent of occurrence (EOO) and an area of occupancy (AOO) of 12 km^2. In the absence of data about population trends, since the species has a restricted range and grows in a fragile habitat it is especially sensitive to environmental changes, so it could easily be impacted by human activity, natural causes, or by catastrophic events. Therefore, according to IUCN (2012, 2017) *A. minoricensis* could be proposed as Vulnerable (VU) under criterion D2. However there is a need for more detailed information on population size and trends to strengthen the assessment.

***Paratypes*—Spain.**—BALEARIC ISLANDS: Minorca: Ciutadella de Menorca, Algaiarens [beach], towards Ses Fontanelles, communities of annual plants in stabilized coastal sands [40.040°, +3.914°], 8 m, 15 Apr 2004, *P. Fraga 55/2004* (P. Fraga pers. herb.!). Ciutadella de Menorca, Alzinar d'Alforí, towards es Milocar, communities of annual plants in stabilized sand dunes [40.034°, +3.975°], 52 m, 15 Apr 2014, *P. Fraga & L. Sáez LS 7477* (L. Sáez pers. herb.!); ibidem, 12-V-2015, *P. Fraga s.n.* (SEV 287206!).

Aira hercynica Romero Zarco, M.Á.Ortiz & L.Sáez, sp. nov. TYPE: SPAIN. Ciudad Real, Fuencaliente, lane left side from road N420, km 105, pasture in dehesa (open forest) of *Quercus suber* and *Quercus pyrenaica*, 816 m a.s.l. [30.438°, –4.330°], 18 June 2018, *R. Berjano & J.A. Mejías* 16/18 BO (holotype: SEV 287404!; isotype: SEV 28740!).

Annual. **Fertile stems** 9–46.5 cm long, erect. **Leaf blade** 10–70 × 0.3–1.0 mm, usually convolute. **Ligule** 1.5–4 mm long, triangular to oblong-lanceolate, subacute. **Panicle** 3–13.5 × 2–7 cm, lax; panicle branches up to 8.9 cm long, patent to erect-patent; pedicel (0.5–)1–13(–15) mm long, 0.2–4.6(–5) times as long as spikelet, usually flexuose not abruptly thickened at apex; apical thickening of pedicel conical-subcylindrical, approximately twice as long as wide. **Spikelets** 2.3–3.3 mm long, with 2 hermaphrodite

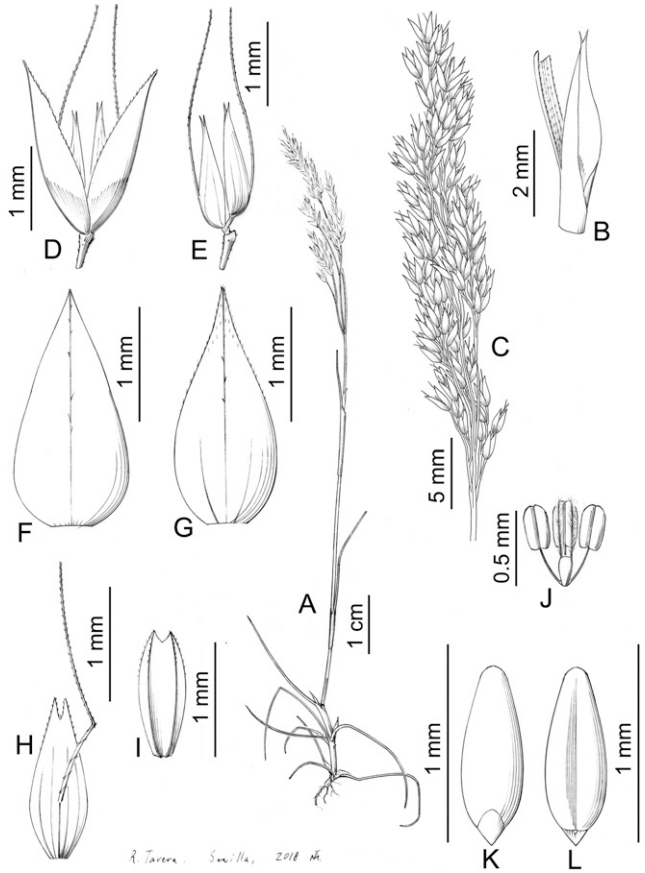


FIG. 5. *Aira minoricensis*. A. Habit. B. Ligule. C. Inflorescence. D. Spikelet. E. Spikelet without the glumes. F. Lower glume. G. Upper glume. H. Lemma. I. Palea. J. Pistil and stamens. K. Caryopsis (abaxial). L. Caryopsis (adaxial). Drawn by R. Tavera (A–B and D–J) and L. Sáez (C). SEV 287206.

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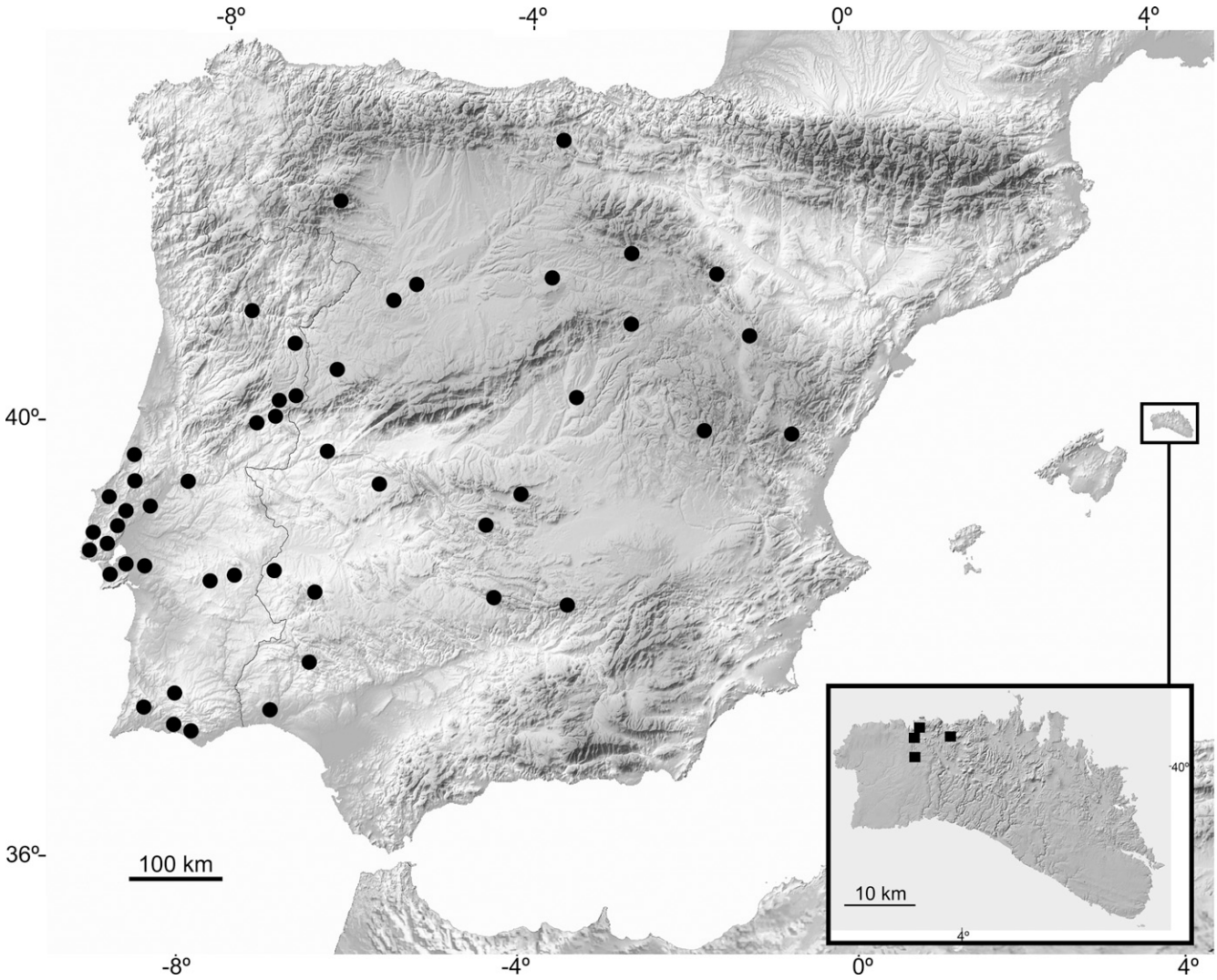


FIG. 6. Geographical distribution of *Aira hercynica* (dots) and *A. minoricensis* (squares).

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awned florets or the lower one unawned (often in the same panicle), laterally compressed. **Glumes** equal to slightly unequal, ovate-lanceolate, obtuse-acuminate or subacute, scabridulous on keel, hyaline to stramineous, shiny; lower glume 2.3–3.0 × 1.3–1.9 mm, 1-veined; upper glume 2.3–3.0 × 1.3–1.8 mm, 3-veined. **Lemma** 1.8–2.6 × 0.5–0.7 mm, bifid at apex, red-brownish, with a dorsal geniculate awn 2.2–3.7 mm long, arising from lower 1/4–1/3 of lemma. **Callus** with hairs (0.1–) 0.2–0.4 mm long. **Palea** 1.4–2.0 × 0.4–0.6 mm, 2-toothed at apex. **Lodicules** ca. 0.3 mm, lanceolate, 2-lobed. **Anthers** (0.6–) 0.7–0.8(–0.9) mm long. **Caryopsis** 1.0–1.3 × ca. 0.3 mm, oblong-elliptic, longitudinally sulcate on adaxial face, glabrous. Figures 3C, 7.

Distribution and Habitat—The species is distributed mainly in the central plateau of the Iberian Peninsula and its surrounding mountains, extending its presence to the west on the coast of Portugal and southwest to the coast of the Huelva province. *Aira hercynica* generally grows in grassland, open scrub, and field margins. It is found on a variety of bedrock types: schist, granite, limestone, and usually on sandy soils in a wide altitudinal range (from near sea level to 1260 m a. s. l). The Guadalquivir River Basin seems to act as a geographic barrier between *A. hercynica* and *A. uniaristata*, probably due to its clayey and marly substrates. However, near coastal areas where sandy substrates predominate, *A. uniaristata* is found on the right side of the river in Doñana National Park, near the southwest limit of *A. hercynica* distribution. No mixed populations have been detected.

Etymology—The specific epithet refers to the Hercynian cratonic block also known as the Iberian Massif, because the distribution of the new species is largely coincident with this area.

Phenology—The flowering period occurs from mid-April to early June, and fruiting time from May to mid-July.

Conservation—*Aira hercynica* is a widely distributed species in the Iberian Peninsula (Fig. 6) where it is found in a variety of open (and sometimes disturbed) habitats over a wide altitudinal range. The species has extent of occurrence (EOO) of 289,921 km^2 and an area of occupancy (AOO) of 184 km^2. There are no significant threats to its future survival. Therefore the species could be proposed as Least Concern (LC) according to IUCN (2012, 2017) criteria.

Additional Specimens Examined—See Supplemental File (Sáez et al. 2019).

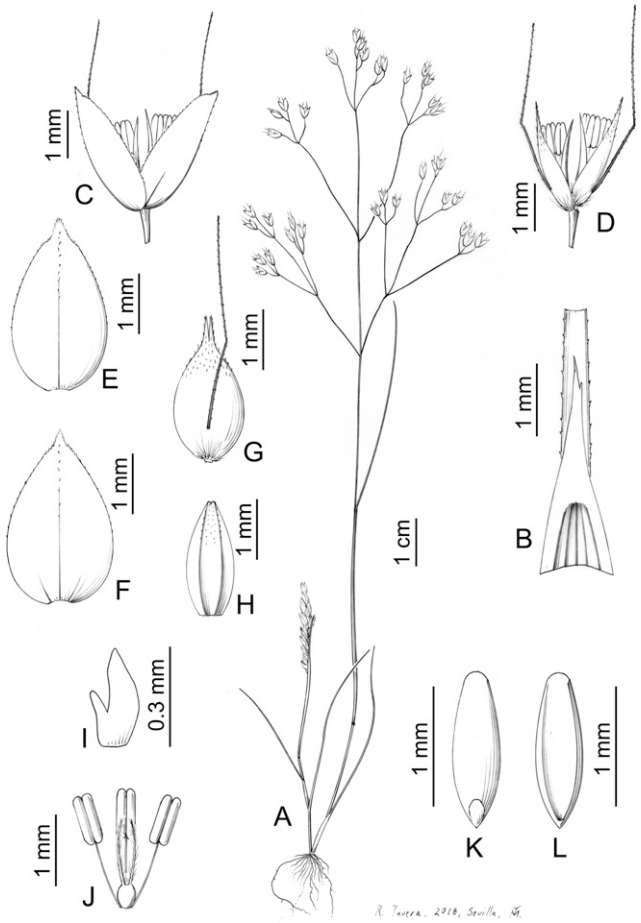


FIG. 7. *Aira hercynica*. A. Habit. B. Ligule. C. Spikelet. D. Spikelet without the glumes. E. Lower glume. F. Upper glume. G. Lemma. H. Palea. I. Lodicule. J. Pistil and stamens. K. Caryopsis (abaxial). L. Caryopsis (adaxial). Drawn by R. Tavera. SEV 287365.

IDENTIFICATION KEY FOR *AIRA* SPECIES IN THE IBERIAN PENINSULA AND THE BALEARIC ISLANDS

1. Panicle rather dense; glumes 3–3.6 mm . . . . . *A. praecox*
1. Panicle very lax to rather dense, but then glumes 1.8–2.2 mm 2
2. Pedicel 2–19 mm, (1–)2–9 times as long as spikelet; spikelets 1.5–2.5 mm 3
3. Spikelets 1.9–2.1 mm; anthers 0.40–0.65 mm; glumes obtuse . . . . . *A. tenorei*
3. Spikelets 1.9–2.5 mm; anthers 0.20–0.35 mm; glumes acuminate . . . . . *A. elegans*
2. Pedicel 0.5–13(–15) mm, (1/4–)1/2–5(–8) times as long as spikelet; spikelets 1.6–3.5 mm 4
4. Pedicel abruptly thickened at apex, with a conspicuous annular widening . . . . . *A. cupaniana*
4. Pedicel not abruptly thickened at apex, without an annular widening 5
5. Panicle usually narrow even in anthesis; spikelets 1.9–2.2 mm long . . . . . *A. minoricensis*
5. Panicle usually lax; spikelets 2.2–3.5 mm long 6
6. Anthers 0.2–0.5(–0.6) mm long; apical thickening of pedicel claviform, about as long as wide . . . . . *A. caryophyllea*
6. Anthers (0.6–)0.7–1.6 mm long; apical thickening of pedicel claviform or not 7
7. Anthers (0.6–)0.7–0.8(–0.9) mm long; apical thickening of pedicel conical-subcylindrical, twice as long as wide . . . *A. hercynica*
7. Anthers 1–1.6 mm long; apical thickening of pedicel claviform, about as long as wide . . . . . *A. uniaristata*

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AUTHOR CONTRIBUTIONS

Sáez performed data collection and analysis, herbarium voucher revision, distribution maps, illustration, and manuscript writing. López-Alvarado performed data analysis and manuscript writing. Fraga performed data collection and manuscript review. Berjano and Ortiz performed data collection and analysis, chromosome counts, plant growing, and manuscript review. Romero-Zarco supervised the taxonomical and nomenclatural decisions, performed data collection and analysis, herbarium voucher revision, manuscript writing, and Supplemental File editing.

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